INCH-POUND

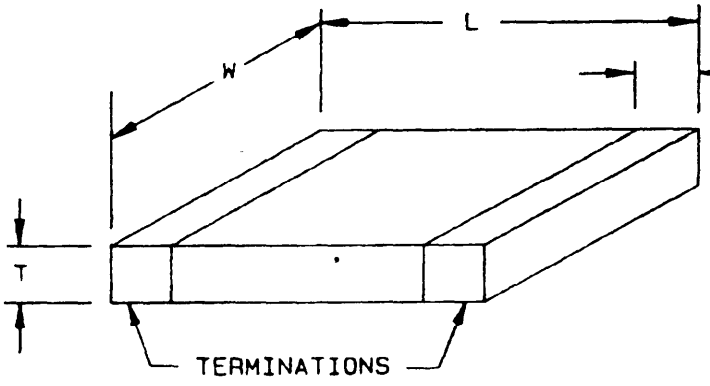
MIL-C-55681/1C
23 April 1992
SUPERSEDING
MIL-C-55681/1B
24 July 1990

MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, MULTIPLE LAYER, FIXED, UNENCAPSULATED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY, STYLES CDR01, CDR02, CDR03, AND CDR04

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55681.



Inches	mm
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76
.050	1.27
.055	1.40
.080	2.03
.125	3.18
.180	4.57

Dimensions				
			T	
Style	L ±.015	W ±.015	Min	Max
CDR01	.080	.050	.022	.055
CDR02	.180	.050	.022	.055
CDR03	.180	.080	.022	.080
CDR04	.180	.125	.022	.080

FIGURE 1. Styles CDR01, CDR02, CDR03, and CDR04 capacitors.

(C) denotes changes

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. For style CDR01, termination width shall be 0.010 inch, minimum; uncovered band width shall be 0.030 inch, minimum.
4. Dimensions and tolerances are for bare chips. For solder coated terminations, for style CDR01 add .020 inch (0.51 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerance. For solder coated terminations for styles CDR02, CDR03, and CDR04, add .025 inch (0.64 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerance.

FIGURE 1. Styles CDR01, CDR02, CDR03, and CDR04 capacitors - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case type: Multilayer, unencapsulated, monolithic.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Marking: In accordance with MIL-C-55681.

TABLE I. CDR01, CDR02, CDR03, and CDR04 capacitors characteristics.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-tempterature limits	Rated voltage
	pF			volts,dc
CDR01BP100B---	10	J, K	BP	100
CDR01BP120BJ--	12	J	BP	"
CDR01BP150B---	15	J, K	BP	"
CDR01BP180BJ--	18	J	BP	"
CDR01BP220B---	22	J, K	BP	"
CDR01BP270BJ--	27	J	BP	"
CDR01BP330B---	33	J, K	BP	"
CDR01BP390BJ--	39	J	BP	"
CDR01BP470B---	47	J, K	BP	"
CDR01BP560BJ--	56	J	BP	"
CDR01BP680B---	68	J, K	BP	"
CDR01BP820BJ--	82	J	BP	"
CDR01BP101B---	100	J, K	BP	"
CDR01B-121B---	120	J, K	BP, BX	"
CDR01B-151B---	150	J, K	BP, BX	"
CDR01B-181B---	180	J, K	BP, BX	"
CDR01BX221B---	220	K, M	BX	"
CDR01BX271BK--	270	K	BX	"
CDR01BX331B---	330	K, M	BX	"
CDR01BX391BK--	390	K	BX	"
CDR01BX471B---	470	K, M	BX	"
CDR01BX561BK--	560	K	BX	"
CDR01BX681B---	680	K, M	BX	"
CDR01BX821BK--	820	K	BX	"
CDR01BX102B---	1,000	K, M	BX	"
CDR01BX122BK--	1,200	K	BX	"
CDR01BX152B---	1,500	K, M	BX	"
CDR01BX182BK--	1,800	K	BX	"
CDR01BX222B--	2,200	K, M	BX	"
CDR01BX272BK--	2,700	K	BX	"
CDR01BX332B---	3,300	K, M	BX	"
(C) CDR01BX392AK--	3,900	K	BX	50
(C) CDR01BX472A---	4,700	K, M	BX	50
CDR02BP221B---	220	J, K	BP	100
CDR02BP271BJ--	270	J	BP	100
(C) CDR02BX392BK--	3,900	K	BX	100
(C) CDR02BX472B---	4,700	K, M	BX	100
CDR02BX562BK--	5,600	K	BX	100
CDR02BX682B---	6,800	K, M	BX	100
CDR02BX822BK--	8,200	K	BX	100
CDR02BX103B---	10,000	K, M	BX	100
CDR02BX123AK--	12,000	K	BX	50

1/ See footnote at end of table.

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TABLE I. CDRO1,CDRO2,CDRO3,and CDRO4 capacitors characteristics - Continued.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			volts,dc
CDRO2BX153A---	15,000	K, M	BX	50
CDRO2BX183AK--	18,000	K	BX	50
CDRO2BX223A---	22,000	K, M	BX	50
CDRO3BP331B---	330	J, K	BP	100
CDRO3BP391BJ--	390	J	BP	100
CDRO3BP471B---	470	J, K	BP	100
CDRO3BP561BJ--	560	J	BP	100
CDRO3BP681B---	680	J, K	BP	100
CDRO3BP821BJ--	820	J	BP	100
CDRO3BP102B---	1,000	J, K	BP	100
CDRO3BX123BK--	12,000	K	BX	100
CDRO3BX153B---	15,000	K, M	BX	100
CDRO3BX183BK--	18,000	K	BX	100
CDRO3BX223B---	22,000	K, M	BX	100
CDRO3BX273BK--	27,000	K	BX	100
CDRO3BX333B---	33,000	K, M	BX	100
CDRO3BX393AK--	39,000	K	BX	50
CDRO3BX473A---	47,000	K, M	BX	50
CDRO3BX563AK--	56,000	K	BX	50
CDRO3BX683A---	68,000	K, M	BX	50
CDRO4BP122BJ--	1,200	J	BP	100
CDRO4BP152B---	1,500	J, K	BP	100
CDRO4BP182BJ--	1,800	J	BP	100
CDRO4BP222B---	2,200	J, K	BP	100
CDRO4BP272BJ--	2,700	J	BP	100
CDRO4BP332B---	3,300	J, K	BP	100
CDRO4BX393BK--	39,000	K	BX	100
CDRO4BX473B---	47,000	K, M	BX	100
CDRO4BX563BK--	56,000	K	BX	100
CDRO4BX823AK--	82,000	K	BX	50
CDRO4BX104A---	100,000	K, M	BX	50
CDRO4BX124AK--	120,000	K	BX	50
CDRO4BX154A---	150,000	K, M	BX	50
CDRO4BX184AK--	180,000	K	BX	50

1/ Complete type designation shall include additional symbol(s) to indicate capacitance tolerance, rated voltage, and voltage-temperature characteristic, (as requied), termination finish, and failure rate level.

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NOTES:

CONCLUDING MATERIAL

Custodian:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - AS, OS
Air Force - 17

User activities:
Navy - MC, SH
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-1807)

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